

Electrospun Microbial-Encapsulated Composite-Based Plasticized Seed Coat for Rhizosphere Stabilization and Sustainable Production of Canola (*Brassica napus* L.)

Zahid Hussain,^{†,‡,§} Muhammad Ali Khan,^{†,‡} Farasat Iqbal,[§] Muhammad Raffi,^{||} and Fauzia Yusuf Hafeez^{*,‡}

[†]Department of Biosciences, COMSATS University Islamabad (CUI), Islamabad, Islamabad Capital Territory 45550, Pakistan

[§]Interdisciplinary Research Centre in Biomedical Materials (IRCBM), COMSATS University Islamabad (CUI), Lahore Campus, Lahore, Punjab 54000, Pakistan

^{||}Department of Materials Engineering, National Institute of Lasers and Optronics (NILOP), Islamabad, Islamabad Capital Territory 45650, Pakistan

Supporting Information

ABSTRACT: Plant-growth-promoting bacteria show promises in crop production; nevertheless, innovation in their stable delivery is required for practical use by farmers. Herein, the composite of poly(vinyl alcohol)/poly(vinylpyrrolidone) plasticized with glycerol and loaded with the microbial consortium (*Bacillus subtilis* plus *Serratia marcescens*) was fabricated and engineered onto canola (*Brassica napus* L.) seed via electrospinning. Scanning electron microscopy showed that the biocomposite is a one-dimensional membrane, which encapsulated microbes in a multilayered nanostructure, and their interfacial behavior between microorganism and seed is beneficial for safer farming. A universal testing machine and thermogravimetric analysis demonstrated that the biocomposite holds sufficient thermomechanical properties for stable handling and practical management. A spectroscopic study resolved the living hybrid-polymer structure of the biocomposite and proved the plasticizing role of glycerol. A swelling study supports the degradation of the biocomposite in the hydrophilic environment as a result of the leaching of the plasticizer, which is important for the sustained release of microbial cells. A shelf life study supported that the biocomposite seed coat placed a threshold level of microbes [$5.675 \pm 0.48 \log_{10}$ colony forming units (CFU)/seed] and maintained their satisfactory viability for 15 days at room temperature. An antifungal and nutrient-solubilizing study supported that the biocomposite seed coat could provide opportunities to biocontrol diseases and improve nutrient acquisition by the plant. A pot study documents the better performance of the biocomposite seed coat on seed germination, seedling growth, leaf area, plant dry biomass, and root system. A chemical and microbial study demonstrated that the biocomposite seed coat improved the effectiveness of the bioinoculant in the root-soil interface, where they survive, flourish, and increase the nutrient pool status. In particular, this study presents advances in the fabrication of the biocomposite for encapsulation, preservation, sustained release, and efficacious use of microorganisms onto seeds for precision farming.

KEYWORDS: microbial consortium, plasticized polymers, electrospinning, encapsulation, shelf life, seed coating, soil properties, canola growth

1. INTRODUCTION

Crops that contain high lipids in their seeds are considered oilseed crops. Canola (*Brassica napus* L.) is an important crop for the production of edible oil, livestock meal, and biofuel products.¹ There is a rising demand for canola consumption, but their production has suffered a setback as a result of the presence of fungal diseases, unavailability of nutrients, and environmental stresses.² Multinational corporations are investing in the production of chemical fertilizers/pesticides for rapid crop growth; nevertheless, their uncontrolled or frequent applications raised various concerns in terms of pollution, salinization, eutrophication, and biological destruction.³ During past few decades, plant-growth-promoting bacteria (PGPB) have shown promise as biostimulants, biopesticides, and biofertilizers. There is a growing need to explore beneficial microorganisms for sustainable crop production and to substitute inputs of synthetic chemicals.^{4,5}

Farming with PGPB to bolster crops has been investigated using various liquid/solid-carrier-based inocula (biopriming, pelleting, microencapsulation, etc.), albeit with pros and cons. Seed application has the advantage over soil application because less microbial mass and time is needed for farming.⁶ Furthermore, microencapsulation is a promising technology that brings advantages in terms of protection, shelf life, handling, and controlled release.⁷ To date, several polymers have been employed for cell encapsulation⁸ using various methods (viz., spray drying, freeze drying, extrusion, and emulsion) for various applications, particularly in the food sector.⁹ Nevertheless, these methods suffered major short-

Received: November 23, 2018

Revised: February 22, 2019

Accepted: April 10, 2019

Published: April 15, 2019

comings in agro-settings: (a) adhere bioinoculants to seeds, (b) high fabrication cost, (c) upscaling at industrial scale, (d) encapsulation of multiple strains, (e) insufficient microbial survival, and (f) uncertainty in root colonization.^{10,11} As a result of patent rights, scarce reports have been published,¹² and therefore, the development of novel inoculation for easy use by farmers and targeted agricultural application is an important issue in the seed industry.

Electrospinning is a simple, economical, and eco-friendly approach to fabricate solid nanofibers (30–1000 nm) from various polymeric solutions using strong electric field produced by dc high voltage as the main driving force.¹³ The unique properties of nanofibers (viz., high surface/volume ratio, tunable porosity, sustained drug release, and high mechanical/thermal stability) combined with their high production rate make this process promising for various industrial applications, such as tissue engineering, cosmetics, wound dressing, catalysis, packaging, and drug delivery.¹⁴ In terms of drug delivery, nanofibers could be used for encapsulation and stable delivery of bacterial strains,¹⁵ spores,¹⁶ yeast cells,¹⁷ mammalian cells,¹⁸ nanoparticles,¹⁹ antibiotics, and plasmids.²⁰ Electrospun nanofibers have recently gained interest in the agriculture industry to provide a new delivery mechanism (e.g., seed coating and food packaging) for agrochemicals for targeted applications. The flexibility in the selection of material and alteration in electrospun collector setup provides an opportunity to fabricate and coat novel biomaterials onto desired surfaces.²¹

Herein, poly(vinyl alcohol) (PVA) and poly(vinylpyrrolidone) (PVP) have been selected as composite encapsulating material as a result of their good biocompatibility, electrospinnability, and biodegradability.^{22,23} Glycerol (Gly) was used as a plasticizer to influence the performance (viz., preserving, mechanical, encapsulation, and sustained release) of materials as a result of the presence of hydroxyl (–OH) groups. The plasticization of polymers²⁴ and engineering of composite macromolecules at nanoscale present opportunities to develop superior biomaterials for multifaceted applications.²⁵ The choice of *Bacillus subtilis* NH-100 and *Serratia marcescens* FA-4 is about combining chemistries (siderophores, cyclic lipopeptide, and nutrient mobilizer) for multiple agricultural purposes, viz., biostimulants, biofertilizers, biopesticides, and biofortification of crops, as well-documented under laboratory and field conditions (bioinoculant was applied by means of soil drenching) in our previous studies.^{26–30} The application of microbial consortium over single inoculation could be beneficial for reducing the harmful impact of stress on plant growth. To boost agronomic practices, the present investigation sought to combine functional properties of the microbial consortium and structural advantages of plasticized polymers in the form of a novel electrospun biocomposite membrane. Furthermore, an electrospun biocomposite was engineered to coat onto canola seeds to place a sufficient density of PGPB to stabilize the rhizosphere and promote sustainable crop production.

2. MATERIALS AND METHODS

2.1. Materials and Microorganisms. PVA (M_w of 110 kDa, 98% hydrolyzed), PVP (M_w of 1300 kDa), and Gly (M_w of 200 g/mol) were purchased from Sigma-Aldrich. Canola [*B. napus* L. cv. Faisal canola (RBN-03060)] seeds were procured from the Oilseed Research Institute of Ayub Agricultural Research Institute (AARI). *B. subtilis* NH-100 (EU627167) and *S. marcescens* FA-4 (KC935385)

were obtained from Pakistan Collection of Microbial Cell Culture (PCMC). *Macrophomina phaseolina*, *Rhizoctonia solani*, and *Fusarium oxysporum* were used as test fungal strains and were provided by the Fungal Pathology Laboratory of Crop Disease Research Institute (CDRI). Media used for bacterial culturing were nutrient broth/agar and Luria–Bertani (LB) agar. Media used for fungal culturing was potato dextrose agar (PDA). The phosphorus solubilization activity was evaluated on Pikovskaya agar medium (amended with 0.1% tricalcium phosphate). All of these media were prepared and used according to the instructions of the manufacturer (Oxoid, Ltd.).

2.2. Production of the Electrospun Biocomposite Seed Coat. To obtain an appropriate amount of bacterial biomass for encapsulation, colonies of *B. subtilis* and *S. marcescens* were separately grown on nutrient broth in 50 mL Falcon tubes in the incubator shaker (IRMECO GmbH) at 100 rpm and 30 °C for 35 ± 5 h until a bacterial population reached up to 10^9 colony forming units (CFU)/mL [optical density (OD) of 595 nm]. The microbial cells were sedimented by centrifugation (centrifuge 5430 R, Eppendorf) at 5000 rpm and 5 °C for 10 min. The obtained biomass was washed twice with saline solution (0.85% NaCl), dried by an inverting tube for 1.5 h at room temperature, and then employed for dispersion preparation.

2.2.1. Preparation of Biopolymeric Dispersions. PVA (10%, w/v), PVP (15%, w/v), and Gly (18%, v/v) solutions were prepared in deionized water (18 MΩ-cm) under magnetic stirring (C-MAC hot plate) for 10 min at 80 °C and then under room temperature until complete hydration of the polymer. PVA/PVP (1:1) solution was prepared by blending an equal amount of PVA and PVP solutions. Further PVA/PVP (1:1) solution was plasticized with various concentrations of Gly to make PVA/PVP/Gly (3:3:1, 3:3:2, and 3:3:3) solutions. Biopolymeric dispersions were prepared by separately adding an aliquot of *B. subtilis*, *S. marcescens*, and *B. subtilis* plus *S. marcescens* in the control (3:3:2 PVA/PVP/Gly) dispersion. The surface tension of prepared solutions was determined using a Sigma tensiometer. The schematic representation of the solution preparation was shown in Figure 1A.

2.2.2. Fabrication of Electrospun Nanofibers. Production of nanofibers was carried out at room temperature (~ 20 °C) using an electrospinning setup consisting of a high voltage dc power supply (50

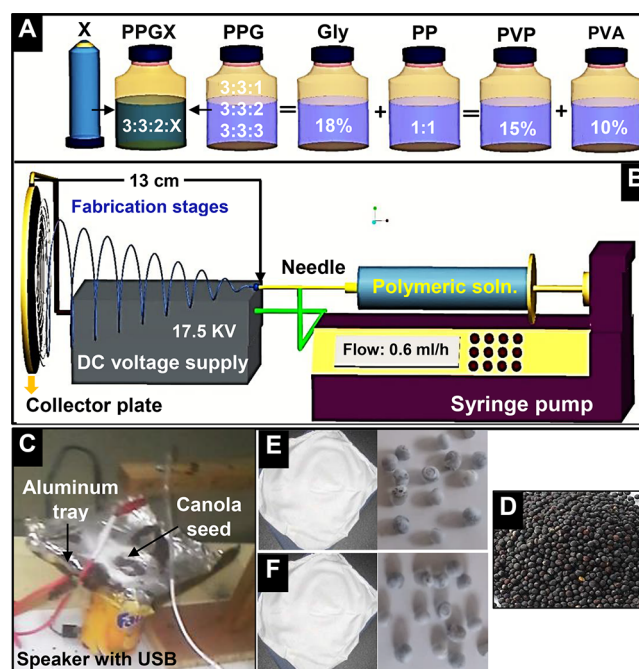


Figure 1. Schematic representation of the (A) solution preparation, (B) electrospinning setup and process to prepare nanofibers, (C) seed-vibrating setup, (D) control canola seed, (E) composite-coated canola seed, and (F) biocomposite-coated canola seed.

kV), 1 mL plastic syringe fitted with a 24-gauge metallic needle, a syringe pump with a flow rate regulator, and an aluminum-foil-wrapped circular copper collector plate (diameter of 20 cm). All prepared dispersions were horizontally discharged at a flow rate of 0.6 mL/h toward the collector plate. A dc high voltage of 17 ± 1 kV was operated at a fixed distance (13 cm) between the needle (positive electrode) and grounded collector (negative electrode). This induced a charge on the liquid droplet surface following Taylor cone formation, launching of the polymer jet, electrospray thinning, whipping instability, and solidification of the fibrous mat on the collector plate.³¹ Electrospun samples were detached, cut into pieces (according to the study objective), and stored in the dark at ambient temperature (~ 20 °C). The resultant 3:3:2 PVA/PVP/Gly, PVA/PVP/Gly/*B. subtilis*, PVA/PVP/Gly/*S. marcescens*, and PVA/PVP/Gly/*B. subtilis*/*S. marcescens* nanofiber seeds were designated as PPG (or composite), PPGB, PPGS, and PPGBS (or biocomposite). The schematic representation of the electrospinning setup is shown in Figure 1B.

2.2.3. Electrospun Coating on Canola Seeds. Electrospun nanofibers were coated onto canola seeds using the electrospinning setup as discussed above, with some modification in the electrospun direction, spinneret, and collector plate. In brief, electrospinning was performed vertically using syringes fitted with a needle through a 15 cm fluid drip pipe. A self-made aluminum tray plate (where seeds were placed) fixed with a sound speaker was used as a conductive collector substrate to collect the charged fibers. A high-beat instrumental tone was played by connecting the universal serial bus (USB) to the audio port of the sound speaker to rotate the canola seeds. Production of nanofibers from the needle was deposited onto vibrated seeds, and corresponding PPG-coated seeds and PPGBS-coated seeds were designated as composite-coated seeds and biocomposite-coated seeds. The weight of seeds before and after coating was taken to determine the amount of applied biomaterial. The photographic images of the electrospun coating setup are shown in Figure 1C. The seed-coating video can be found as Video S1 of the Supporting Information.

2.3. Qualitative Characterization and Measurement. Surface morphology of samples [PVA, PVP, PVA/PVP, PPG (3:3:1, 3:3:2, 3:3:3), PPGB, PPGS, and PPGBS] was observed under scanning electron microscopy (SEM, Tescan Vega-3 LMU) in secondary electron mode at an accelerating voltage of 10–15 kV. ImageJ software [National Institutes of Health (NIH), Bethesda, MD, U.S.A.] was used to reconstruct the SEM image to measure size distributions and porosity related to samples. The elemental quantification of samples (PPG and PPGBS) was investigated using energy-dispersive X-ray spectroscopy (EDS, Oxford Instrument) at an accelerating voltage of 20 keV using carbon (C, CaCO_3), oxygen (O, SiO_2), phosphorus (P, GaP), sulfur (S, FeS_2), and nitrogen (N, not defined) as standards.

2.3.1. Mechanical and Thermal Study. Mechanical properties, such as tensile strength (σ), elongation at break (ϵ), and elastic modulus (E) of rectangular strips (width, 20 mm; length, 50 mm), of samples (PVA, PVP, PVA/PVP, PPG, and PPGBS) were analyzed using a universal testing machine (UTM, ATS-FAAR) equipped with a 10 N load cell under a cross-head speed of 10 mm/min and gauge length of 25 mm. Thermal stability of samples was studied using thermogravimetric analysis (TGA, Mettler Toledo). In brief, the approximate weight (≈ 10 mg) of each sample was separately pressed in a standard sealed aluminum pan and heated from 25 to 400 °C at the heating rate of 10 °C/min under a nitrogen atmosphere (50 mL/min).

2.3.2. Spectroscopic Study. The chemical composition, change in functional group, and chemical interaction associated with prepared samples (PVA, PVP, PVA/PVP, PPG, and PPGBS) were investigated using Fourier transform infrared spectroscopy (FTIR, Nicolet 6700) and Raman spectroscopy (μ Ramboss, Dangwoo). In brief, the FTIR spectrum was collected in transmission mode from the wavenumber range of 400–4000 cm^{-1} using 32 scans at 4 cm^{-1} spectral resolution. The Raman spectrum was collected in a wavenumber range of 400–1800 cm^{-1} with 4 cm^{-1} spectral resolution using helium–cadmium as

a laser radiation source. The OriginPro8 program was used to generate spectral figures independent of the numerical range, and the ChemSketch program was used to make schematic molecular interactions present in samples.

2.4. In Vitro Assessment of the Biocomposite Membrane.

To determine the swelling index, pieces of composite and biocomposite of equal weight (50 mg, W_i) were incubated at 37 °C within 10 mL of saline solution. After the predetermined time (1, 2, 3, 4, 5, 6, and 7 h), samples were removed and water on their surface was air-dried and then reweighed (W_d). Experiments were performed in triplicate and swelling (%) was calculated using eq 1.³²

$$\text{swelling (\%)} = W_i - W_d / W_d \times 100 \quad (1)$$

2.4.1. Survival Study. The survival rate of entrapped microbes in the biocomposite membrane and the biocomposite seed coat with respect to their storage time was periodically (1, 5, 10, 15, 20, and 25 days) evaluated as $\log_{10}$ colony forming units ($\log_{10}$ CFU/g or $\log_{10}$ CFU/seed) and was compared to the survival rate of free microbes in biopolymer dispersion ($\log_{10}$ CFU/mL). These experiments were performed in triplicate using the spread plate serial dilution method on nutrient agar plates. Colonies were counted (after incubation at 30 °C until 48 h) by a colony counter using plates containing 30–300 colonies,³³ followed by digital processing of the image using the OpenCFU 3.9.0 program.³⁴ Viability was calculated using eq 2.

$$\text{viability} = (\text{colonies number} \times \text{dilution factor}) / \text{inoculum volume} \quad (2)$$

2.4.2. Functional Study. The antifungal potential of the biocomposite membrane (diameter of 7 mm, on the third day of storage) was assessed in triplicate against *M. phaseolina*, *R. solani*, and *F. oxysporum* by the dual culture method on a PDA plate. Inhibition of fungal mycelium was monitored after the sixth day of incubation at 30 °C. To determine P solubilizing potential, the biocomposite membrane (diameter of 5 mm, on the third day of storage) was placed in the middle of Pikovskaya agar medium. The presence of a hollow zone around the biocomposite was monitored after the fourth day of incubation at 30 °C, which indicated phosphate solubilizing ability. Further, to report the qualitative physiological performance of the electrospun-recovered microorganism, *S. marcescens* was quantified for red pink colony production on LB agar plates at 28 °C,³⁵ whereas *B. subtilis* was quantified on the basis of creamy colony production on LB agar plates containing 5% NaCl at 37 °C.³⁶

2.5. Pot Experiment. The pot experiment was conducted following a completely randomized design in the net house (Green Valley Nursery, Islamabad, Pakistan) under natural conditions (light/dark photoperiod, 14/10 h; day temperature, ~ 28 °C; and night temperature, ~ 15 °C) with three treatments: (a) uncoated seed, (b) composite-coated seed, and (c) biocomposite-coated seed. The experimental soil used for cultivation was sandy loam, and their microbial and physicochemical properties were discussed in Table 2. The soil was filled in three plastic germination trays (42 square cells per tray). Each square cell (diameter of 7 cm and depth of 9 cm) contains 110 ± 10 g of soil. About 150 seeds from each treatment in three replications were sown 2.80 ± 0.20 cm deep in soil containing pots (5 seeds/square cell) on the fifth day of the electrospun coating.

2.5.1. Growth Parameters. Data on agronomic traits of 15 plants from each treated group at random per replicate were measured using 14-day-old seedlings to find out the efficacy of biocomposite coating. In brief, root length and shoot length was measured in millimeters using a meter scale. Plant fresh and dry biomass was determined by weighing (mg). Germination percentage, number of leaves, lateral root branches, and root hairs per plant were counted numerically. The total leaf surface index was determined using ImageJ software. The seed germination rate (eq 3) and plant vigor index (eq 4) were carried according to Kan et al.³⁷

$$\text{germination rate} = (\text{germinated seeds} / \text{total seeds}) \times 100 \quad (3)$$

$$\text{vigor index} = (\text{root length} + \text{shoot length}) \times \% \text{ germination} \quad (4)$$

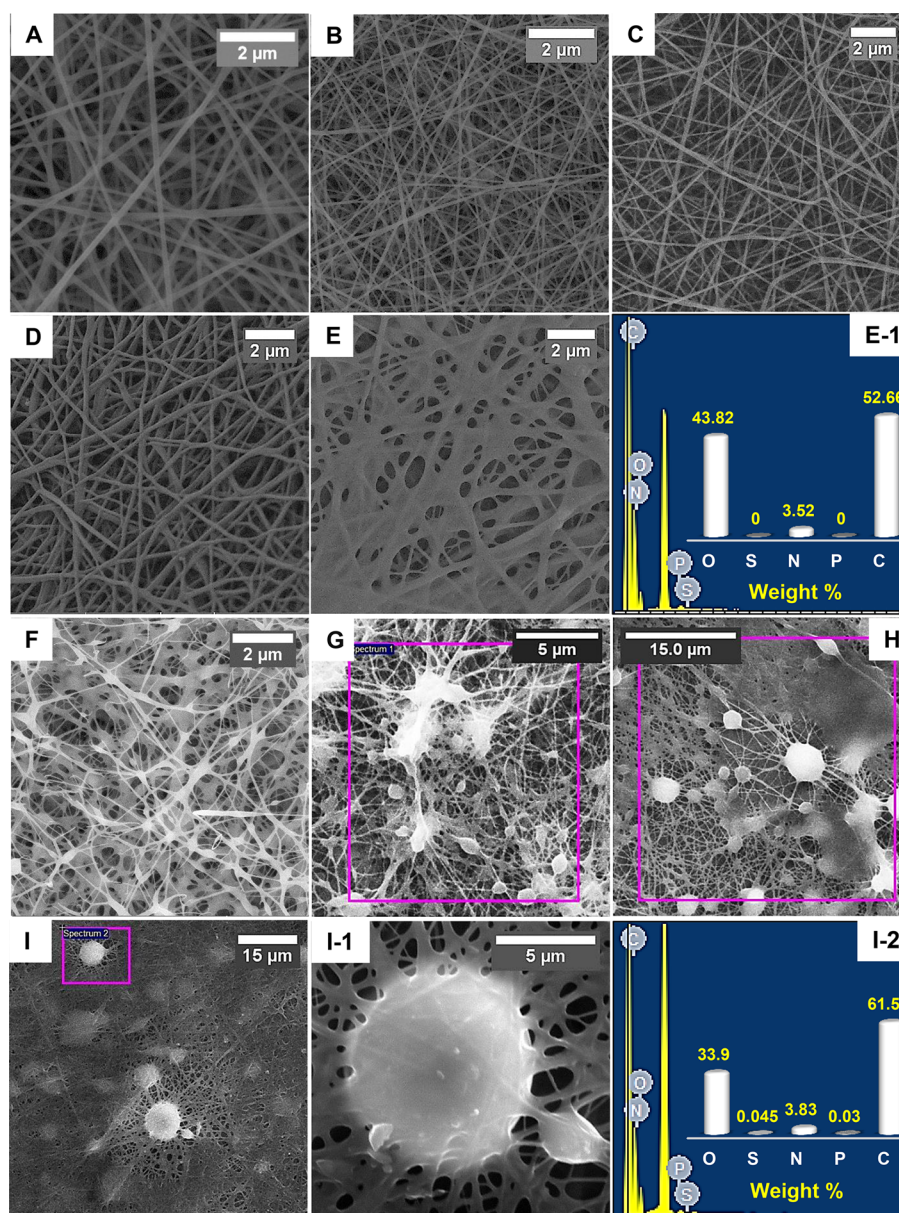


Figure 2. SEM micrographs showing morphology of nanofibers of (A) PVA, (B) PVP, (C) PP, (D) PPG-3:3:1, (E) PPG-3:3:2 or composite, (F) PPG-3:3:3, (G) PPGB, (H) PPGS, (I) PPGBS or biocomposite. (I-1) Closeup view of the entrapped microorganism. (E-1) EDS micrograph of the composite. (I-2) EDS micrograph of the biocomposite.

2.5.2. Microbiological and Physicochemical Characterization of Soil. A total of 1 g of rhizosphere (soil under the influence of roots) associated with each treatment group (selected random per replicate) was washed with saline solution (pH 7.0) and used to enumerate the microbial population by the spread plate serial dilution method on the nutrient agar plate. Furthermore, rhizospheric samples were sent to the Soil & Water Testing Laboratory for Research, Rawalpindi, Pakistan, to determine its physical (pH, clay, sand, silt, and organic matter) and chemical (total N, P, K, and Zn) properties.

2.6. Statistical Analysis. Qualitative studies were performed under identical settings of the instrument, and representative images were shown. Data values of CFU were log-transformed before analysis. All quantitative experimental results were analyzed using the Statistix 8.1 program and denoted as the mean $\pm$ standard deviation (SD). The least significant difference (LSD) test was applied to evaluate mean significant differences ($p \leq 0.05$).

3. RESULTS AND DISCUSSION

3.1. Macro- and Microscopic Morphologies of the Biocomposite Seed Coat. A canola seed is 1.5 mm in diameter, oval in shape, black in color, and 0.42 ± 0.03 g in weight (Figure 1D). The direct seed coating with nanofibers appeared as the white membrane onto the individual seed surface during 20 ± 5 min of electrospinning. Afterward, coated seeds begin to clump, which may due to the adhesive property of the composite membrane (discussed in section 3.2). Upon visual observation, the nanofiber coating gives a white color to the seeds and helps in the distinction between coated and non-coated seeds. There was no macroscopic difference between the composite seed coat (Figure 1E) and the biocomposite seed coat (Figure 1F), and fiber coating was found to offer better layer coverage and adhesion onto the seed. Unlike Krishnamoorthy et al.³⁸ and Castañeda et al.³⁹ with the seed coating approach on a static seed using a simple

polymeric composition, herein, we use a sound-vibrating approach using novel biocomposite chemistries for homogeneous electrospun coating. The consolidated value of the fiber coating weight per seed is about 0.016 ± 0.015 g; thus, this technological coating reduced the consumption of biomaterials compared to bioprimering methods. Moreover, nanofibers, being long-chained and adhesive particularly at the time of fabrication, can attach at the solid surface of seed sites probably through a mechanical interlocking effect.

A SEM micrograph demonstrated that PVA (Figure 2A), PVP (Figure 2B), and PVA/PVP (Figure 2C) solutions were spun into non-woven, smooth, beadless, and elongated fibers with the nanostructured diameter distribution in the range of 100 ± 10 nm. Various ratios of Gly were investigated to select suitable composition based on morphometric results. SEM micrographs of PPG-3:3:1 (Figure 2D) demonstrated that the addition of Gly increased fiber diameter (350 ± 20 nm). SEM micrographs of PPG-3:3:2 (Figure 2E) demonstrated that the increase in Gly percentage increased fiber diameter (370 ± 10 nm), promoted a uniform interconnected spongiform structure, and dispersion of additional glycerol over main fibrous layers. This indicated that Gly improves the superior cross-linked network through hydrogen bonding between $-OH$ groups on PVA and carbonyl ($C=O$) groups on PVP at various sites. A SEM micrograph of PPG-3:3:3 (Figure 2F) demonstrated that a higher concentration of Gly acted as an antiplasticizer, which did not allow the fiber to dry, migrate out of the matrix, and produce a non-uniform merged fibrous structure.⁴⁰ In particular, the composite membrane is a one-dimensional thin nanomaterial membrane (PPG-3:3:2) selected as a control as a result of its interconnected nanostructure, surface smoothness, and reduced porosity. Furthermore, EDS of the composite membrane showed the presence of O, C, and N as main elemental components, which reflect the quantitative presence of PVA/PVP/Gly (Figure 2E-1).

SEM micrographs of the PPGB membrane showed many coccus-shaped spots (diameter of 1.5 ± 0.5 μ m), indicating the encapsulation of *B. subtilis*, which is a Gram-positive and non-motile bacillus (Figure 2G). SEM micrographs of PPGBS membrane showed many larger coccus spots (diameter of 5.5 ± 3.5 μ m), indicating encapsulation of two or more cells of *S. marcescens* (Figure 2H), which is a Gram-negative and motile bacillus. SEM micrographs of the biocomposite membrane (PPGBS) showed many smaller spots (diameter of 2.0 ± 0.1 μ m), indicating encapsulation of *B. subtilis*, and many larger spots (diameter of 6.0 ± 3.6 μ m), indicating encapsulation/immobilization of *S. marcescens* (Figure 2I). Furthermore, close observation of the immobilized spot in the biocomposite membrane demonstrated that encapsulation of microorganisms caused local widening of fibers (Figure 2I-1). This indicated that internal spaces of biocomposite fibers allowed the microorganism to survive, multiply, and produce metabolites at a minute amount within the fibrous nanostructure (further discussed in section 3.4). EDS showed that O, C, N, P, and S are the elemental components of the biocomposite, which reflect the quantitative presence of the PVA/PVP/Gly/microbial consortium (Figure 2I-2).

In summary, bacteria in the biocomposite dispersion supposed to cause a slight change in the membrane structure might as a result of the action of microbial surfactant products,⁴¹ which lower the surface tension ($64.6 \pm 0.08 \rightarrow 58.8 \pm 0.05$ mN/m) properties of dispersion compared to the

control dispersion, requiring less charge density ($18 \rightarrow 16$ kV) to prompt the formation of thin nanofibers.⁴² In particular, electrospinning is an economical, eco-friendly, and precise technique to fabricate a one-dimensional thin biocomposite membrane. The biocomposite encapsulates and integrates the microbial consortium within the multilayered nanostructured network. The modification in the electrospun collector setup allowed for the coating of sufficient numbers of canola seeds with the biocomposite membrane. The interfacial behavior of the biocomposite between the microorganism and seed could be beneficial for the safer management and protection of microorganisms.

3.2. Thermal Stability of the Biocomposite Seed Coat.

Thermogram results (Figure 3) demonstrated that PVA

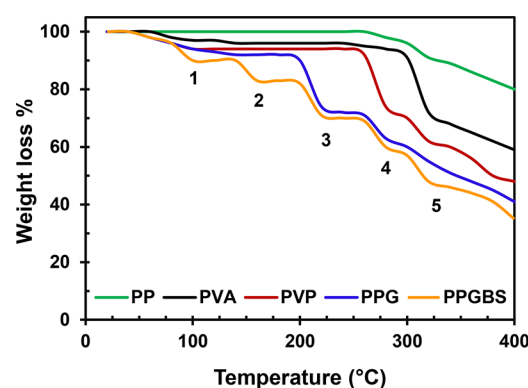


Figure 3. Weight change percentage value of electrospun nanofibers of PVA, PVP, PP, PPG, and PPGBS as a function of the temperature ($^{\circ}$ C) obtained from TGA under the nitrogen atmosphere. The biocomposite membrane showed weight loss in five stages: first stage corresponded to the dehydration of loosely bound water; the second stage represented the thermal signatures of encapsulated bacterial components; the third stage corresponded to the evaporation of the plasticizer; the fourth stage corresponded to the melting of the crystalline domain of polymers; and the fifth stage represents the decomposition of the side group of polymers.

showed 24% weight loss at the temperature range of $300\text{--}350$ $^{\circ}$ C, representing the decomposition of the polymer chains. PVP showed 30% weight loss at the temperature range of $250\text{--}320$ $^{\circ}$ C, corresponding to the degradation of the polymer chain. PVA/PVP nanofibers showed a much higher thermal resistance property compared to their counterparts maybe as a result of the strong intermolecular interactions between $-OH$ groups of PVA and $C=O$ groups of PVP, which masked the additional bonding sites. The composite showed weight loss in four stages: The first stage showed 8% weight loss ($25\text{--}150$ $^{\circ}$ C), corresponding to the dehydration of loosely bound water; the second stage showed 19% weight loss ($200\text{--}250$ $^{\circ}$ C), corresponding to the decomposition of the plasticizer;⁴³ the third stage showed 11% weight loss ($250\text{--}300$ $^{\circ}$ C), corresponding to the melting of the crystalline domain of polymers (PVA and PVP); and the fourth stage showed 14% weight loss ($300\text{--}360$ $^{\circ}$ C), representing the decomposition of the side group of polymers. The biocomposite demonstrated a thermal behavior similar to the composite counterpart, except that an additional decomposition step (9% weight loss occurred) has been noticed in the temperature range of $150\text{--}200$ $^{\circ}$ C, which probably represents the thermal signatures of bacterial components.⁴⁴ In particular, thermal results showed that Gly caused the breakdown of crystalline domains

and is interposed between the strand of PVA and PVP polymers in the composite membrane. The biocomposite holds an acceptable level of thermal stability, which is useful for the farming application and the protection of encapsulated microorganisms against environmental stresses.⁴⁵

3.3. Mechanical Stability of Biocomposites. The mechanical results (Table 1) showed a decrease in the tensile

Table 1. Variation in Mechanical Properties of Nanofibrous Samples Obtained from a UTM Equipped with a 10 N Load Cell under a Cross-Head Speed of 10 mm/min

sample	thickness (mm)	tensile strength (MPa)	Young modulus (MPa)	elongation at break (%)
PVA	0.015 ± 0.0005	15.2 ± 0.5	55 ± 4.5	110 ± 10
PVP	0.014 ± 0.0008	13.6 ± 0.2	50 ± 7.5	105 ± 12
PP	0.015 ± 0.0004	24.5 ± 0.6	65 ± 8.0	85 ± 05
PPG	0.016 ± 0.0006	10.8 ± 0.1	40 ± 5.0	140 ± 08
PPGBS	0.017 ± 0.0008	08.3 ± 0.4	45 ± 4.0	145 ± 06

strength and elastic modulus while an increase in percent elongation in the order of PPGBS > PPG > PVP > PVA > PP nanofibers. The transitions in composite properties demonstrated that the presence of Gly reduces hardness and elastic modulus. It gives structural flexibility to the composite membrane by reducing and competing for the intermolecular interaction between polymeric chains of PVA and PVP at various sites.⁴⁶ Upon visual and touching observation, we found that the composite and biocomposite membrane are more adhesive compared to the PVA/PVP fibrous membrane maybe as a result of solubility parameters of the plasticizer, which improved their adhesive strength. In particular, the biocomposite exhibits a sufficient level of mechanical properties, which is attractive for stable handling of microorganisms during practical application. Furthermore, the adhesive property of the biocomposite is important for mechanical and chemisorption coating onto many application sites (e.g., seed surfaces).

3.4. Physicochemical Properties of Biocomposites. Different modes of FTIR-based molecular vibration (stretching, bending, scissor, twisting, and wagging motions) had been studied that verified the presence of major functional groups (OH, CH, CH₂, C=O, C–N, and C=C) and confirmed that polymers do not undergo degradation during the electrospinning. The characteristic PVA bands were observed at 3330 cm^{−1} (OH stretching), 2930 cm^{−1} (CH asymmetric stretching), 1422 cm^{−1} (C–O stretching), 1330 cm^{−1} (CH₂ scissoring), 1091 cm^{−1} (C–O stretching of the acetyl group), 922 cm^{−1} (CH bending), 844 cm^{−1} (CH₂ rocking), and 664 cm^{−1} (C–C stretching). The characteristic PVP bands were observed at 3424 cm^{−1} (OH stretching), 2930 cm^{−1} (CH asymmetric stretching), 1652 cm^{−1} (amide I stretching vibration), and 1291 cm^{−1} (amide III vibration).⁴⁷ The PVA/PVP blend showed an overlapped spectral pattern of their counterparts, except for the presence of 3356 cm^{−1}, which indicates the existence of hydrogen bonding between –OH groups of PVA and C=O groups of PVP. The composite showed the same spectra as PVA/PVP counterparts, except for the shifting of OH stretching (3356 → 3289 cm^{−1}), C–O stretching (1091 → 1036 cm^{−1}) bands and the presence of new bands at 2860 cm^{−1}, which indicate that the –OH groups of Gly competes for hydrogen bonding between PVA and PVP chains at various sites. The biocomposite showed the same

spectral pattern as composite counterparts, except for the change in the intensities of 3289 and 1291 cm^{−1} bands (Figure 4A), which could be the spectral fingerprints of entrapped microbes.⁴⁸

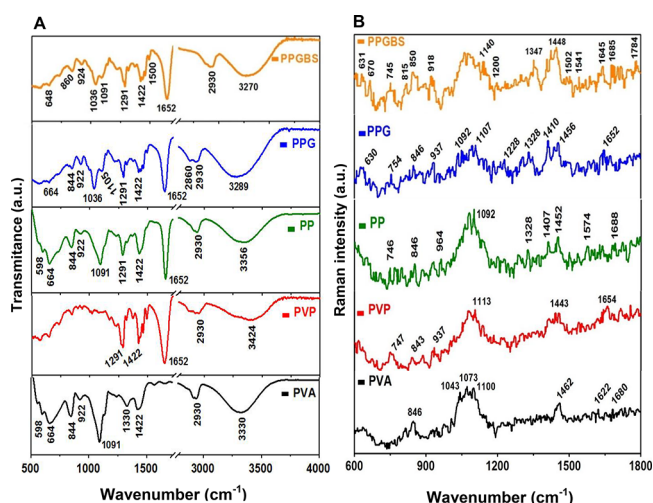


Figure 4. Comparative spectrum results of electrospun nanofibers of PVA, PVP, PP, PPG, and PPGBS obtained by (A) FTIR and (B) Raman spectroscopy.

Different modes of Raman spectral vibration (viz., frequency, polarization, width, and intensity) have been studied to provide information about the composition, amount, and quality of the material. PVA/PVP nanofibers had demonstrated the bands at 1658 cm^{−1} (amide I stretching vibration of PVP), 1445 cm^{−1} (C–O stretching vibration of PVP), 1375 cm^{−1} (OH stretching of PVA), 1115 cm^{−1} (amide III stretching vibration of PVP), 935 cm^{−1} (C–C bonding), and 1174 cm^{−1} (CH stretching of PVA and PVP). The vibrational band at 1092 cm^{−1} in the PVA/PVP composite is assigned to C=O stretching, which confirmed the intermolecular interactions between the –OH groups of PVA and the C=O groups of PVP. In the composite, the reduction in intensity of the 1092 cm^{−1} band showed the decreasing intermolecular interaction between PVA and PVP, whereas an increase in the intensity of 1652 and 1222 cm^{−1} bands showed intensive hydrogen bonding of Gly with polymeric chains of PVA and PVP. The biocomposite showed the same spectral pattern as composite counterparts, except for the change in the intensities of 1460 and 1092 cm^{−1} bands and the presence of many new uncertain peaks (Figure 4B), which could be the spectral fingerprints of microbes.⁴⁹ Qualitative outcomes resolved the living hybrid–polymer structure of the biocomposite membrane.

3.5. Swelling Performance of Biocomposites. The observed weight-swelling percentage values of the composite membrane were 122 ± 4.5, 154 ± 3.0, 179 ± 3.7, 158 ± 4.4, 116 ± 4.0, 94 ± 3.5, and 82 ± 2.0, and the observed swelling percentage values of the biocomposite membrane were 103 ± 5.0, 131 ± 4.5, 159 ± 6.0, 133 ± 4.0, 99 ± 3.5, 83 ± 1.5, and 72 ± 1.0% for the time intervals of 1, 2, 3, 4, 5, 6, and 7 h, respectively (Figure 5). It is inferred that the presence of high numbers of –OH group in Gly is the reason for favorable sorption of saline fluids by the biocomposite membrane at initial hours. Moreover, the swelling percentage of membranes constantly decreases in the hydrophilic environment as a result

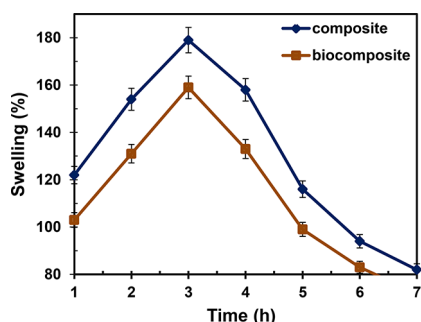


Figure 5. Comparison of the weight-swelling index of the composite membrane and biocomposite membrane as a function of time (h) in saline solution at 37 °C.

of the leaching of Gly (plasticizer), which disorganized cross-linked fibrous chains, and is essential for on-site activation and for the sustained release of microbial cells in the hydrophilic environment.

3.6. Shelf Life of the Biocomposite Seed Coat. *In vitro* survivability demonstrated that the biocomposite membrane holds $8.65 \pm 0.08 \log_{10}$ CFU/g, whereas biopolymeric dispersion holds $9.62 \pm 0.15 \log_{10}$ CFU/mL culturable microbial cells after the first day of their storage at room temperature. On the 25th day, the biocomposite membrane showed significant improvement in the microbial survival rate by $3.9 \pm 0.04 \log_{10}$ units compared to the free microorganism in polymeric dispersion. The higher viscosity of the biopolymeric solution may be one of the detrimental factors that reduced the viability of free microorganisms. Moreover, the highest viable count has been noticed from the fifth day of biocomposite storage ($8.91 \pm 0.06 \log_{10}$ CFU/g), which demonstrated that biocomposites provide a suitable micro-encapsulated environment for constrained growth of encapsulated microorganisms (panels A and B of Figure 6). The

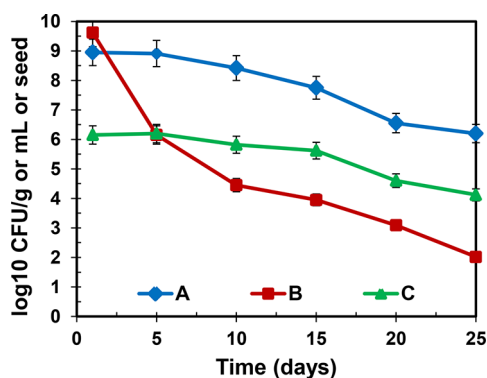


Figure 6. Comparison of the semi-log plot of viable cells of microorganisms as a function of the storage time (days) on a nutrient agar plate for the (A) biocomposite membrane (CFU/g) (B) biopolymeric dispersion (CFU/mL) and (C) biocomposite seed-coat (CFU/seed). Bars represent the mean $\pm$ SD ($n = 3$). Sources of error are the weighing of material, dilution steps, and colony counting steps.

electrospun biocomposite membrane thus presents an improvement in microbial survival compared to unencapsulated microbes. These findings are in line with the reports in which bacterial cells have been found encapsulated at room temperature in the various polymeric systems.^{50,51} Furthermore, *in vitro* survivability demonstrated that the biocomposite seed coat holds $6.15 \pm 0.05 \log_{10}$ CFU/seed on the first day of

storage and $4.12 \pm 0.06 \log_{10}$ CFU/seed at the 25th day of their storage. Because a canola seed is small in size (1.5 mm in diameter), it can be coated with a narrow amount of bioinoculant. The biocomposite seed coat could place the threshold of PGPB ($5.675 \pm 0.48 \log_{10}$ CFU/g) during the first 15 days of their storage at room temperature (Figure 6C); therefore, the use of the biocomposite seed coat within 15 days from the manufactured date is beneficial. Moreover, the interfacial behavior of the biocomposite between the microorganism and seed could be beneficial for the safer management of microbes because it could protect microorganisms against inhibitory exudates present in the seed tegument.

3.7. Functional Performance of the Biocomposite Seed Coat. Fungal pathogens cause serious losses to agricultural crops. Fungicides have been used to control pathogens; nevertheless, their frequent application is responsible for superbug development, and their chemical residues can accumulate in the food products. PGPB emerged as a better subsistent to combat fungal disease.⁵² Composite membranes had not shown antifungal potential, whereas the biocomposite membrane showed variable antifungal potential against *F. oxysporum* (15 ± 0.5 mm; Figure 7A), *R. solani* (13

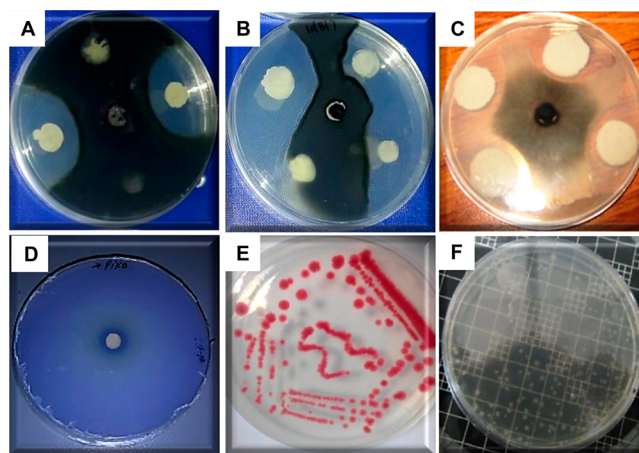


Figure 7. (A) Zone of inhibition potential of the biocomposite against *M. phaseolina*, (B) zone of inhibition potential of the biocomposite against *R. solani*, (C) zone of inhibition potential of the biocomposite against *F. oxysporum*, (D) phosphate solubilizing potential of the biocomposite on Pikovskaya agar medium, (E) representative red pink colonies of *S. marcescens* on LB agar plate at 28 °C, and (F) representative creamy colonies of *B. subtilis* nutrient colonies on LB agar plates containing 5% NaCl at 37 °C.

± 1.0 mm; Figure 7B), and *M. phaseolina* (14 ± 0.6 mm; Figure 7C) as a result of the presence of two superior microbial strains, which could biocontrol plant diseases under *in vitro* and field experiments.^{26,30}

Canola needs a high amount of nutrients than other crops for balanced plant growth. Many soils are rich in nutrients but present in insoluble complexes and, thus, cannot be available to support the plant.⁵³ PGPB is an important component of nutrient mobilization and soil sustainability.⁵⁴ Composite membranes had not shown nutrient solubilizing potential, while the biocomposite membrane solubilizes the insoluble ores of P (Figure 7D). The nutrient solubilizing potential of the biocomposite correlates with the viable presence of the entrapped microorganism, which could provide unique opportunities to improve nutrient acquisition.^{26,28} Further,

colony morphological results of the electrospun-recovered microorganism demonstrated that *S. marcescens* activity produced the pink–red pigment (Figure 7E) and *B. subtilis* colonies produced the creamy pigment (Figure 7F). This support that the functional performance of the entrapped microorganism was not affected by electrospinning and coating processes.

3.8. Role of Plasticizer, Bioeffectors, and Structural Architecture of the Biocomposite. Microscopic, qualitative, and *in vitro* outcomes proved the plasticizing role of Gly on the structural, mechanical, thermal, swelling, and shelf life performances of the biocomposite seed coat. In brief, the plasticizer has the same affinities as the individual polymers forming the composite and, thus, homogeneous distribution throughout the membrane to improve flexibility and control the sustained release performance of the biocomposite. It could protect the bacterium from the rapid dehydration during electrospinning and improve the chemisorption between the biocomposite and microbes, which is important for better encapsulation of microorganisms. Moreover, it could improve adhesion and chemisorption between the biocomposite and seed, which is important for better coverage of seeds with nanofibers. The schematic molecular interactions involved in the composite membrane and biocomposite membrane are illustrated in Figure 8.

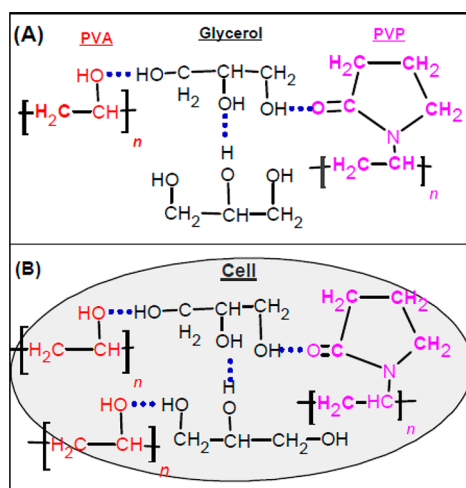


Figure 8. Schematic representation of molecular interactions existing in the (A) composite membrane and (B) biocomposite membrane.

The biocontrol and nutrient solubilizing study proved the bioeffector role of the encapsulated microbial consortium in the biocomposite seed coat. The structural architecture of the biocomposite not only provides a suitable nanoenvironment for microbial survival but also improves the physiological response. It could ensure the placement of the threshold level of the microbial consortium onto the seed for targeted application in the root–soil interface. Moreover, the interfacial behavior of the biocomposite between the microorganism and seed could be beneficial for the safer management of microbes for precision farming. These properties are crucial for preparing ecologically safe and practically applicable bio-stimulants, biofertilizers, and biopesticides for sustainable crop protection and production.⁵⁵

3.9. Practical Performance of the Biocomposite Seed Coat. The high initial density of the biocomposite seed coat is

important from a commercial point of view. The pot results demonstrated that the rate of vegetative growth varied with the coating nature. The biocomposite-coated seed showed an improvement in seedling germination and establishment rate (Figure 9A) and saved up to 40% seeds compared to the

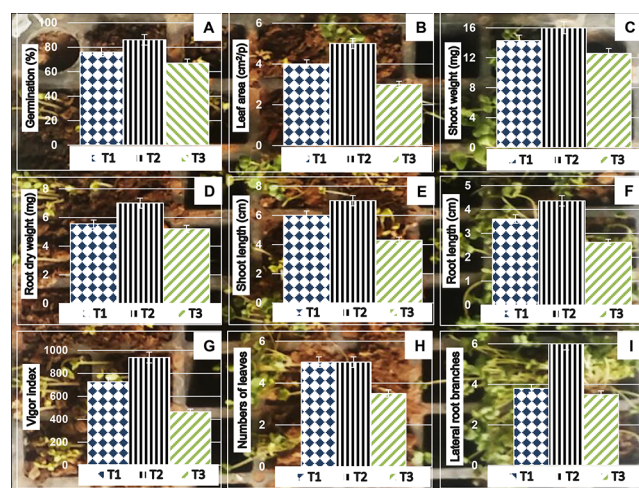


Figure 9. Effect of treatments (T1, composite-coated seed; T2, biocomposite-coated seed; and T3, uncoated seed) on 14-day-old seedling plant traits: (A) seedling germination, (B) leaf surface area, (C) shoot dry weight, (D) root dry weight, (E) shoot length, (F) root length, (G) vigor index, (H) number of leaves, and (I) number of lateral root branches. Values are the average $\pm$ SD, indicated by error bars ($n = 15$; $p < 0.05$).

control seed. The biocomposite coat enhanced the leaf surface area (Figure 9B) and plant dry weight (panels C and D of Figure 9). It promoted the root and shoot length (panels E and F of Figure 9). It improved the vigor establishment of plants (Figure 9G) compared to untreated seeds and composite-coated seeds. The number of leaves per plant remains the same in both the composite seed and biocomposite seed treatments; however, the control plant shows a reduced number and yellowing of some leaves (Figure 9H). The biocomposite seed coat promoted the lateral root branches and the development of root hair (Figure 9I), which are the main factors enhancing crop yields. These results document the satisfactory effects of seed coating on enhancing numerous indicators, such as seed germination, seedling growth, root and shoot growth, leaf area, and dry biomass of seedling.

Functional stabilization of the root–soil interface (rhizosphere) is an important area for plant–microbe interactions and sustained plant growth.⁵⁶ *In vivo* microbial survival analysis supports that the higher population of a culturable bacterial population ($8.50 \log_{10}$ CFU/g) is associated with the rhizosphere of biocomposite seedling compared to the rhizosphere associated with composite seedling ($3.75 \log_{10}$ CFU/g) and untreated seedling ($3.80 \log_{10}$ CFU/g). This demonstrated that the biocomposite seed coat successfully placed the microbes at their root–soil interface. Moreover, the significant difference in the microbial population of a cultivated soil ($1.45 \log_{10}$ CFU/g) and control plant rhizosphere ($3.80 \log_{10}$ CFU/g) demonstrated that exudates produced by the plant roots could recruit and flourish rhizobacteria. The physicochemical study demonstrated that the biocomposite-coated seed presents beneficial effects on the soil chemical properties, such as increasing soil pH and bioavailability of

Table 2. Effect of the Composite Seed Coat and Biocomposite Seed Coat on the Microbiological and Physicochemical Properties of the Rhizosphere

soil sample	microbial properties	physical properties					chemical properties			
	count (log ₁₀ CFU/g)	pH (H ₂ O)	clay (%)	sand (%)	silt (%)	organic C (%)	N (g/kg)	K (mg/kg)	Zn (mg/kg)	P (mg/kg)
cultivation soil	1.45	7.96	14.4	68.6	17.0	1.16	1.35	11.46	0.80	6.5
untreated soil rhizosphere	3.80	7.78	4.3	68.6	17.1	1.88	1.94	11.58	0.90	7.76
composite amendment	3.75	7.65	4.3	68.6	17.1	1.92	1.96	11.58	0.90	7.80
biocomposite amendment	8.5	7.52	14.3	68.5	17.2	2.02	2.15	11.64	1.08	10.6

nutrients, particularly Zn and P, to restore soil fertility for better crop response compared to the composite-coated seed. Moreover, the degradation products of the composite could serve as a provision carbon source for microbial survival and seedling growth (Table 2).

In summary, our results indicate that application of a biocomposite over the seed surface is a superior nano-encapsulation carrier system that is able to improve the effectiveness of the bioinoculant as a result of accurate delivery in the root–soil interface. The better growth performance of seedling and expanded root system demonstrated that delivered the microbial consortium survives, flourishes, and interacts intrinsically with plant roots and improves the nutrient pool status of the rhizosphere. This demonstrated that a high microorganism count and the level of nutrients improved the growth stimulus to the target seedling. There are numbers of reasons for this positive effect, such as decaying cells of microorganisms, mineralization activities of microorganisms, and more microbial activity stimulating root exudation, which, in return, increases the biological and chemical performance of the rhizosphere. In particular, stabilization of chemical and biological processes could assist to strengthen seedlings to cope with stress conditions and counter seed/soil-borne diseases.

On the whole, this study presents advances in the fabrication of a novel electrospun microbial composite-based seed coat for encapsulation, preservation, handling, sustained release, and efficacious use of microorganisms for precision farming. The interfacial behavior of the biocomposite between the microorganism and seed could be beneficial for the safer management of microbes and the placement of a sufficient density of the bioinoculant in the root–soil interface. The biocomposite seed coat contains superior microorganisms, which hold biostimulant, biopesticide, and biofertilizer properties for sustainable production of canola seedlings. The novelty of this green technology is based on designing the nanochemistry of the biocomposite and their coating on sound-vibrated canola seeds to reduce the cost of cultivation. This study could open a potential horizon to coat various small seeds of important crops with the polymer-based bioinoculant formulation. For more comprehensive results, further studies should be directed to assess the effect of seed coatings on canola biofortification and strengthen obtained results under fertigation and stress systems. Moreover, fabrication productivity of the biocomposite seed coat could be burgeoning with improving the industrial-grade electrospinning setup to accelerate the adoption into crop practices.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jafc.8b06505.

Electrospun seed coating (Video S1) (MP4)

■ AUTHOR INFORMATION

Corresponding Author

*Telephone: +92-300-6602614. E-mail: fauzia@comsats.edu.pk, fauzia_y@yahoo.com.

ORCID

Zahid Hussain: 0000-0002-2824-2672

Author Contributions

†Zahid Hussain and Muhammad Ali Khan contributed equally to this work and are co-first authors.

Funding

The authors are thankful to the Deanship of NIOP and IRCBM for providing technical and instrument support for this research article.

Notes

The authors declare no competing financial interest.

■ ABBREVIATIONS USED

PVA, poly(vinyl alcohol); PVP, poly(vinylpyrrolidone); Gly, glycerol; PGPB, plant-growth-promoting rhizobacteria; SEM, scanning electron microscopy; TGA, thermogravimetric analysis; EDS, energy-dispersive X-ray spectroscopy; FTIR, Fourier transform infrared spectroscopy; UTM, universal testing machine; CFU, colony forming units; PDA, potato dextrose agar

■ REFERENCES

- (1) Sharma, M.; Gupta, S. K.; Mondal, A. K. Production and trade of major world oil crops. In *Technological Innovations in Major World Oil Crops*; Gupta, S. K., Ed.; Springer: New York, 2012; Vol. 1, pp 1–15, DOI: 10.1007/978-1-4614-0356-2_1.
- (2) Chattopadhyay, C.; Kolte, S. J.; Waliyar, F. *Diseases of Edible Oilseed Crops*; CRC Press: Boca Raton, FL, 2015.
- (3) Savci, S. Investigation of effect of chemical fertilizers on environment. *APCBEE Proc.* **2012**, 1, 287–292.
- (4) Hafeez, F. Y.; Yasmin, S.; Ariani, D.; ur-Rahman, M.; Zafar, Y.; Malik, K. A. Plant growth-promoting bacteria as biofertilizer. *Agron. Sustainable Dev.* **2006**, 26 (2), 143–150.
- (5) Compant, S.; Duffy, B.; Nowak, J.; Clément, C.; Barka, E. A. Use of plant growth-promoting bacteria for biocontrol of plant diseases: Principles, mechanisms of action, and future prospects. *Appl. Environ. Microbiol.* **2005**, 71 (9), 4951–4959.
- (6) O'Callaghan, M. Microbial inoculation of seed for improved crop performance: Issues and opportunities. *Appl. Microbiol. Biotechnol.* **2016**, 100 (13), 5729–5746.
- (7) John, R. P.; Tyagi, R. D.; Brar, S. K.; Surampalli, R. Y.; Prévost, D. Bio-encapsulation of microbial cells for targeted agricultural delivery. *Crit. Rev. Biotechnol.* **2011**, 31 (3), 211–226.
- (8) de Vos, P.; Lazarjani, H. A.; Poncelet, D.; Faas, M. M. Polymers in cell encapsulation from an enveloped cell perspective. *Adv. Drug Delivery Rev.* **2014**, 67–68, 15–34.

- (9) Solanki, H. K.; Pawar, D. D.; Shah, D. A.; Prajapati, V. D.; Jani, G. K.; Mulla, A. M.; Thakar, P. M. Development of micro-encapsulation delivery system for long-term preservation of probiotics as biotherapeutics agent. *BioMed Res. Int.* **2013**, *2013*, 1–21.
- (10) Schoebitz, M.; López, M. D.; Roldán, A. Bioencapsulation of microbial inoculants for better soil-plant fertilization. A review. *Agron. Sustainable Dev.* **2013**, *33* (4), 751–765.
- (11) Bashan, Y.; de-Bashan, L. E.; Prabhu, S.; Hernandez, J. P. Advances in plant growth-promoting bacterial inoculant technology: Formulations and practical perspectives (1998–2013). *Plant Soil* **2014**, *378* (1–2), 1–33.
- (12) Kim, R.; Kapstein, E. B. *Who Benefits from Intellectual Property Rights for Agricultural Innovation?*; Steward Redqueen, Haarlem, Netherlands, 2015.
- (13) Bhardwaj, N.; Kundu, S. C. Electrospinning: A fascinating fiber fabrication technique. *Biotechnol. Adv.* **2010**, *28* (3), 325–347.
- (14) Persano, L.; Camposeo, A.; Tekmen, C.; Pisignano, D. Industrial upscaling of electrospinning and applications of polymer nanofibers: A review. *Macromol. Mater. Eng.* **2013**, *298* (5), 504–520.
- (15) Zussman, E. Encapsulation of cells within electrospun fibers. *Polym. Adv. Technol.* **2011**, *22* (3), 366–371.
- (16) Spasova, M.; Manolova, N.; Naydenov, M.; Kuzmanova, J.; Rashkov, I. Electrospun biohybrid materials for plant biocontrol containing chitosan and *Trichoderma viride* spores. *J. Bioact. Compat. Polym.* **2011**, *26* (1), 48–55.
- (17) Letnik, I.; Avrahami, R.; Rokem, J. S.; Greiner, A.; Zussman, E.; Greenblatt, C. Living composites of electrospun yeast cells for bioremediation and ethanol production. *Biomacromolecules* **2015**, *16* (10), 3322–3328.
- (18) Canbolat, M. F.; Tang, C.; Bernacki, S. H.; Pourdeyhimi, B.; Khan, S. Mammalian cell viability in electrospun composite nanofiber structures. *Macromol. Biosci.* **2011**, *11* (10), 1346–1356.
- (19) Zhang, C.; Yu, S. Nanoparticles meet electrospinning: Recent advances and future prospects. *Chem. Soc. Rev.* **2014**, *43* (13), 4423–4448.
- (20) Lee, S.; Jin, G.; Jang, J.-H. Electrospun nanofibers as versatile interfaces for efficient gene delivery. *J. Biol. Eng.* **2014**, *8* (1), 30.
- (21) Noruzi, M. Electrospun nanofibres in agriculture and the food industry: A review. *J. Sci. Food Agric.* **2016**, *96* (14), 4663–4678.
- (22) Teodorescu, M.; Bercea, M. Poly(vinylpyrrolidone)—A versatile polymer for biomedical and beyond medical applications. *Polym.-Plast. Technol. Eng.* **2015**, *54* (9), 923–943.
- (23) Muppalaneni, S.; Omidian, H. Polyvinyl alcohol in medicine and pharmacy: A perspective. *J. Dev. Drugs* **2013**, *2* (3), 1–5.
- (24) Snejdrova, E.; Ditttrich, M. Pharmaceutical applications of plasticized polymers. In *Recent Advances in Plasticizers*; Luqman, M., Ed.; IntechOpen: London, U.K., 2012; Vol. 159, pp 23–34, DOI: 10.5772/36543.
- (25) Zucchelli, A.; Focarete, M. L.; Gualandi, C.; Ramakrishna, S. Electrospun nanofibers for enhancing structural performance of composite materials. *Polym. Adv. Technol.* **2011**, *22* (3), 339–349.
- (26) Abaid-Ullah, M.; Hassan, M. N.; Jamil, M.; Brader, G.; Shah, M. K.N.; Sessitsch, A.; Hafeez, F. Y. Plant growth promoting rhizobacteria: An alternate way to improve yield and quality of wheat (*Triticum aestivum*). *Int. J. Agric. Biol.* **2015**, *17*, 51–60.
- (27) Hassan, M. N.; Namood-e-Sahar; Shah, S. Z.; Afghan, S.; Hafeez, Y. F. Suppression of red rot disease by *Bacillus* sp. based biopesticide formulated in non-sterilized sugarcane filter cake. *BioControl* **2015**, *60* (5), 691–702.
- (28) Shakeel, M.; Rais, A.; Hassan, M. N.; Hafeez, F. Y. Root associated *Bacillus* sp. improves growth, yield and zinc translocation for basmati rice (*Oryza sativa*) varieties. *Front. Microbiol.* **2015**, *6*, 1286.
- (29) Hassan, M. N.; Afghan, S.; Hafeez, F. Y. Biological suppression of sugarcane red rot by *Bacillus* spp. under field conditions. *J. Plant. Pathol.* **2012**, *94* (2), 325–329.
- (30) Sarwar, A.; Hassan, M. N.; Imran, M.; Iqbal, M.; Majeed, S.; Brader, G.; Sessitsch, A.; Hafeez, F. Y. Biocontrol activity of surfactin A purified from *Bacillus* NH-100 and NH-217 against rice bakanae disease. *Microbiol. Res.* **2018**, *209*, 1–13.
- (31) Theron, S. A.; Zussman, E.; Yarin, A. L. Experimental investigation of the governing parameters in the electrospinning of polymer solution. *Polymer* **2004**, *45* (6), 2017–2030.
- (32) Ganesh, M.; Aziz, A. S.; Ubaidulla, U.; Hemalatha, P.; Saravanakumar, A.; Ravikumar, R.; Peng, M. M.; Choi, E. Y.; Jang, H. T. Sulfanilamide and silver nanoparticles-loaded polyvinylalcohol–chitosan composite electrospun nanofibers: Synthesis and evaluation on synergism in wound healing. *J. Ind. Eng. Chem.* **2016**, *39* (22), 127–135.
- (33) Meynell, G. G.; Meynell, E. *Theory and Practice in Experimental Bacteriology*; Cambridge University Press, New York, 1965; pp 288.
- (34) Geissmann, Q. OpenCFU, a new free and open-source software to count cell colonies and other circular objects. *PLoS One* **2013**, *8* (2), No. e54072.
- (35) Hardjito, L.; Huq, A.; Colwell, R. R. The influence of environmental conditions on the production of pigment by *Serratia marcescens*. *Biotechnol. Bioprocess Eng.* **2002**, *7* (2), 100–104.
- (36) Dobrovolskaya, T. G.; Skvortsov, I. N.; Lysak, L. V. *Methods for the Identification and Isolation of Soil Bacteria*; Moscow State University Press: Moscow, Russia, 1990.
- (37) Kan, G.; Zhang, W.; Yang, W.; Ma, D.; Zhang, D.; Hao, D.; Hu, Z.; Yu, D. Association mapping of soybean seed germination under salt stress. *Mol. Genet. Genomics* **2015**, *290* (6), 2147–2162.
- (38) Krishnamoorthy, V.; Elumalai, G.; Rajiv, S. Environment friendly synthesis of polyvinylpyrrolidone nanofibers and their potential use as seed coats. *New J. Chem.* **2016**, *40* (4), 3268–3276.
- (39) Castañeda, L. M.; Genro, C.; Roggia, I.; Bender, S. S.; Bender, R. J.; Pereira, C. N. Innovative Rice Seed Coating (*Oryza sativa*) with Polymer Nanofibres and Microparticles Using the Electrospinning Method. *J. Res. Updates Polym. Sci.* **2014**, *3* (1), 33–39.
- (40) Lin, C. A.; Ku, T. H. Shear and elongational flow properties of thermoplastic polyvinyl alcohol melts with different plasticizer contents and degrees of polymerization. *J. Mater. Process. Technol.* **2008**, *200* (1–3), 331–338.
- (41) Sarwar, A.; Brader, G.; Corretto, E.; Aleti, G.; Abaidullah, M.; Sessitsch, A.; Hafeez, F. Y. Qualitative analysis of biosurfactants from *Bacillus* species exhibiting antifungal activity. *PLoS One* **2018**, *13* (6), No. e0198107.
- (42) Reneker, D. H.; Yarin, A. L. Electrospinning jets and polymer nanofibers. *Polymer* **2008**, *49* (10), 2387–2425.
- (43) Basiak, E.; Lenart, A.; Debeaufort, F. How glycerol and water contents affect the structural and functional properties of starch-based edible films. *Polymers* **2018**, *10* (4), 412.
- (44) Lee, J.; Kaletunç, G. Evaluation of the heat inactivation of *Escherichia coli* and *Lactobacillus plantarum* by differential scanning calorimetry. *Appl. Environ. Microbiol.* **2002**, *68* (11), 5379–5386.
- (45) Fung, W. Y.; Yuen, K. H.; Liong, M. T. Agrowaste-based nanofibers as a probiotic encapsulant: Fabrication and characterization. *J. Agric. Food Chem.* **2011**, *59* (15), 8140–8147.
- (46) Mohsin, M.; Hossin, A.; Haik, Y. Thermal and mechanical properties of poly(vinyl alcohol) plasticized with glycerol. *J. Appl. Polym. Sci.* **2011**, *122* (5), 3102–3109.
- (47) Mondal, D.; Mollick, Md. M. R.; Bhowmick, B.; Maity, D.; Bain, M. K.; Rana, D.; Mukhopadhyay, A.; Dana, K.; Chattopadhyay, D. Effect of poly(vinylpyrrolidone) on the morphology and physical properties of poly(vinyl alcohol)/sodium montmorillonite nanocomposite films. *Prog. Nat. Sci.* **2013**, *23* (6), 579–587.
- (48) Maquelin, K.; Kirschner, C.; Choo-Smith, L.-P.; van den Braak, N.; Endtz, H. P.; Naumann, D.; Puppels, G. J. Identification of medically relevant microorganisms by vibrational spectroscopy. *J. Microbiol. Methods* **2002**, *51* (3), 255–271.
- (49) Stöckel, S.; Kirchhoff, J.; Neugebauer, U.; Rösch, P.; Popp, J. The application of Raman spectroscopy for the detection and identification of microorganisms. *J. Raman Spectrosc.* **2016**, *47* (1), 89–109.
- (50) Nagy, Z. K.; Wagner, I.; Suhajda, Á.; Tobak, T.; Harasztsos, A. H.; Vigh, T.; Söti, P. L.; Pataki, H.; Molnár, K.; Marosi, G.

Nanofibrous solid dosage form of living bacteria prepared by electrospinning. *eXPRESS Polym. Lett.* **2014**, *8* (5), 352–361.

(51) López-Rubio, A.; Sanchez, E.; Sanz, Y.; Lagaron, J. M. Encapsulation of living bifidobacteria in ultrathin PVOH electrospun fibers. *Biomacromolecules* **2009**, *10* (10), 2823–2829.

(52) Singh, J. S.; Pandey, V. C.; Singh, D. P. Efficient soil microorganisms: A new dimension for sustainable agriculture and environmental development. *Agric., Ecosyst. Environ.* **2011**, *140* (3–4), 339–353.

(53) Boyles, M.; Peeper, T.; Stamm, M. *Great Plains Canola Production Handbook*; Kansas State University Agricultural Experiment Station and Cooperative Extension Service: Manhattan, KS, 2012; pp 6–18.

(54) Malusà, E.; Ciesielska, J. Biofertilizers: A resource for sustainable plant nutrition. In *Fertilizer Technology*; Sinha, S., Pant, K. K., Bajpai, S., Eds.; Studium Press LLC: Houston, TX, 2015; Vol. I: Synthesis, pp 282–319.

(55) Campos, E. V.; De Oliveira, J. L.; Fraceto, L. F.; Singh, B. Polysaccharides as safer release systems for agrochemicals. *Agron. Sustainable Dev.* **2015**, *35* (1), 47–66.

(56) Morgan, J. B.; Connolly, E. L. Plant–soil interactions: Nutrient uptake. *Nat. Educ. Knowl.* **2013**, *4* (8), 2.

■ NOTE ADDED AFTER ASAP PUBLICATION

This paper published ASAP on April 24, 2019 with errors in the Figure 6 caption. The corrected version reposted to the Web on April 25, 2019.